Assignment 2 | Multilevel Models for Educational Data

Executed report • original INVALSI spreadsheet and dictionary • ML/REML fits with two random-effect levels

1. Research objective and data source

The assignment studies associations between standardized mathematics achievement and pupils' backgrounds using a three-level educational structure: pupils nested within class sections, nested within school campuses (Italian grade G05). Data: **INVALSI_data_MAT_2324.xlsx**, with definitions and missing-value codes verified using **MAT_05-2023-24_datadescription.xlsx**. The outcome is **WLE_MATH_200**, a continuous item-response-theory-based math score. Neither **math_score_raw** nor **math_score_percentile** was included as a predictor, in line with the instructions.

Analyses were independently re-executed with the supplied original spreadsheets using Python, statsmodels and a fully executed Jupyter notebook. Reported values originate from the new fits, not from prior repository outputs.

2. Cleaning, missingness and representativeness

Quantity	Observed value	Meaning
Original student records	16,244	One row per observed pupil
Complete-case records	9,623 (59.24%)	6,621 excluded (40.76%)
School campuses	400	Observed school identifiers after cleaning
Nested classes	757	Distinct school-class identifiers
Classes per school	357 campuses with 2; 43 with 1	Two variance components can be distinguished
Mean math score	197.800 (SD 37.392)	Among 9,623 complete cases

Cleaning was based on the **supplied dictionary**: 9 = missing for origin and nursery/kindergarten attendance; 99 = missing for parent education and occupation; gender = 9, year = 9, and birth month = 99 are missing codes when present. Importantly, **month = 9 means September, not missing**. Non-numeric ESCS values were coerced to missing; complete cases on all selected variables were retained. The parent education code 9 is a valid educational category, while 99 represents missing.

Geographic area code	Retained / total	Retention rate
1 - North-West	2,060 / 3,164	65.11%
2 - North-East	2,029 / 3,263	62.18%
3 - Centre	1,691 / 3,022	55.96%
4 - South	2,323 / 3,797	61.18%
5 - South and Islands	1,520 / 2,998	50.70%

Selection is substantial and differs across geographic areas. The uncleaned dataset has a mean WLE_MATH_200 of 195.073 versus 197.800 among complete cases. This comparison suggests nonrepresentative case deletion; missing-at-random assumptions were not established.

3. Variable classes and model structure

Level	Variables	Treatment
Student (Level 1)	Gender, birth month, birth year, origin, nursery/kindergarten, parental education and occupation, ESCS_student	Categorical codes entered as factor indicators; individual ESCS continuous
Class (Level 2)	ESCS_class, CLASS_ANONYMIZED	Class ESCS as fixed covariate; nested random intercept
School (Level 3)	SCHOOL_ANONYMIZED	Campus random intercept
Contextual	GeoArea_5	5 geographical area indicators; code 1 reference

Region_Code was inspected during data cleaning but excluded from the fitted formula because the model already accounts for broader geography using GeoArea_5. The original source has sampling weights but this mixed-effects analysis is **unweighted**; estimates should not be interpreted as automatically design-weighted national estimates.

4. Model specification and estimation

Mixed model: $Y_{ijk} = X_{ijk}\beta + u_k + v_{jk} + e_{ijk}$, where u_k is a school random intercept, v_{jk} is a class-within-school random intercept, and e_{ijk} is an individual residual; these components are assumed independent and Gaussian, with variances σ^2_{school}, σ^2_{class} and σ^2_e. The full fixed-effect model includes every variable listed in Section 3, plus macro-geography. Birth month and year are coded as factors, matching the dictionary.

To evaluate whether measured family and socioeconomic variables improve fit, a reduced model was estimated using gender, birth month, birth year, origin and geography only. **Both nested models were compared under maximum likelihood (ML)**, using identical student records and identical random structure. For final reported parameter estimates the full model was refitted using restricted maximum likelihood (REML). All three optimization runs converged.

ML specification	Log likelihood	AIC	BIC
Reduced	-47,810.074	95,670.149	95,849.447
Full	-47,416.701	94,963.403	95,429.577

Likelihood ratio test: chi-square(40) = 786.746, p = 2.5e-139. The full fixed-effects specification fits substantially better than the reduced specification. The improvement is statistical, not evidence that every individual new coefficient is significant or causally important.

5. Fixed-effects results and substantive interpretation

Fixed effect (reference shown)	Estimate	SE	p-value
Female vs male	-10.197	0.671	3.649e-52
First-generation immigrant vs native	-7.493	2.405	0.001834
Second-generation immigrant vs native	-6.622	1.338	7.434e-07
No nursery vs yes	-3.199	0.753	2.13e-05
No kindergarten vs yes	-1.464	2.155	0.4969
Student socioeconomic index	+3.302	0.930	0.0003823
Class socioeconomic index	+1.800	1.477	0.2229
South and Islands vs North-West	-5.513	2.285	0.01582
Father education: PhD vs primary	+15.940	5.199	0.002169

Fixed effect (reference shown)	Estimate	SE	p-value
Mother education: PhD vs primary	+15.512	5.319	0.003543

Holding the other recorded covariates and model random intercepts constant, female pupils (code 2) have approximately **10.20 score units lower** expected achievement than male pupils (code 1). Compared with native pupils, first- and second-generation immigrant categories are lower by approximately **7.49** and **6.62** points, respectively. Nonattendance at nursery is associated with about **3.20** fewer points relative to attendance. The kindergarten attendance coefficient is not significant at the 5% level.

An increase of one unit in ESCS_student is associated with **3.30 points higher** expected score (p=0.00038); ESCS_class is positive (+1.80) but not statistically significant (p=0.223). The South and Islands geographic category (5) scores **5.51 points lower** than North-West (1) after adjustments (p=0.0158). Many higher parental education categories have positive associations relative to primary schooling, although each category contrast depends on sparse/reference counts and multiple-testing considerations.

Birth month was treated as categorical, avoiding a forced linear numerical interpretation; births later in the calendar year show generally more negative adjusted estimates relative to January, consistent with a possible relative-age association but not a causal demonstration.

6. Random effects, clustering and ICC

Variance parameter	REML estimate	Interpretation
Between-school variance	66.234	Shared school differences
Between-classes-within-school variance	122.458	Additional shared class differences
Within-class residual variance	1,027.225	Pupil-specific variation remaining
ICC: different classes, same school	0.0545 (5.45%)	Correlation attributable to common school
ICC: same class	0.1552 (15.52%)	Correlation attributable to school + class

The positive estimated variances establish that the observed data support separate school and nested-class components. The intraclass correlation is calculated under the independent random-intercept variance decomposition; it does not mean that school identity *causes* 5.45% of student performance.

7. Diagnostics and limitations

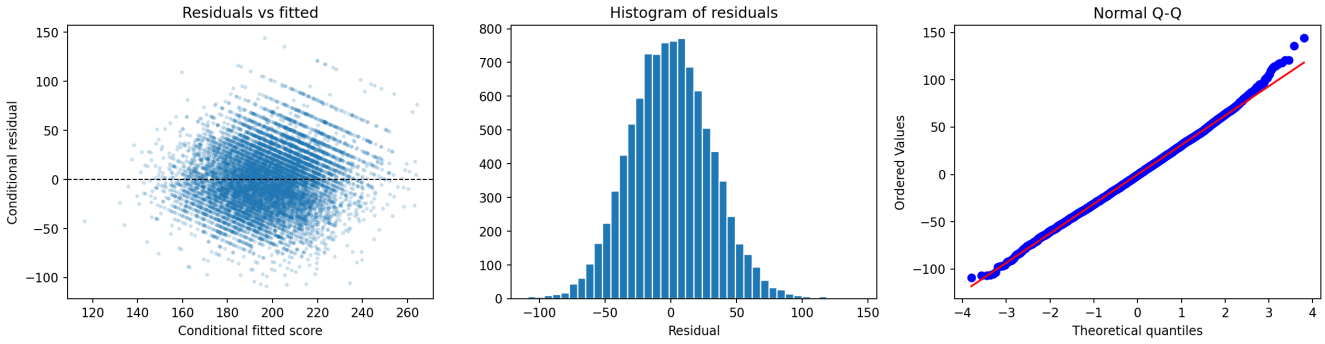


Figure 1. Conditional residual-versus-fitted plot, histogram, and normal Q-Q plot from the final REML fit. Diagonal streaks reflect the partially discrete IRT-derived outcome scores.

Conditional residuals have approximately zero mean and SD 31.13; the histogram is broadly centered and symmetric, while the Q-Q plot shows departures at the tails, especially the high end. The residual-versus-fitted plot shows a dense, banded cloud; visual assessment does not guarantee homoscedasticity or exact Gaussian tails. Interpretation of fixed-effect p-values is conditional on the working model and large-sample approximations.

• Complete-case deletion removes 40.76% of the original sample and shows unequal geographic retention; estimates could suffer selection bias.

• Unequal student/school sampling weights and the stratified survey design were not incorporated in this unweighted mixed-effects fit.

• An observational cross-sectional dataset cannot identify causal effects of gender, family SES, migration origin, teaching quality or early education.

• Random slopes, alternative missingness approaches and survey-weighted/hierarchical methods are reasonable extensions, not conclusions from the fitted model.

8. Conclusion

The updated analysis supports a three-level Gaussian mixed-effects approach for INVALSI grade-5 mathematics. It **separately** identifies nonzero school and classroom variance and finds evidence that demographic, early-education, family and individual socioeconomic covariates improve model fit. The individual ESCS effect is positive and significant, while the class ESCS effect is not individually significant. Because missingness is substantial and the survey sampling design is simplified, conclusions must be interpreted as conditional associations for the analysed complete cases, not causal or nationally design-weighted estimates.

Reproducibility and submission materials

Sources: Instructions_2.pdf (Giulia Vannucci, 26 Nov 2025); INVALSI_data_MAT_2324.xlsx; MAT_05-2023-24_datadescription.xlsx. Code: executed assignment2.ipynb; extracted fit tables, model comparison and diagnostic figure under results/. Required submission folder: Assignments / Assignment2 / name_surname_assignment2 (PDF, code, individual/group note).